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# On-farm rainwater and crop management for improving productivity of rainfed areas

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**Abstract**

Scarcity of water is a critical limitation to adoption of modern technology for increasing productivity of traditional rainfed rice growing areas of eastern Madhya Pradesh, India. The shortage of water results from uneven distribution of rains, significant gaps between rain events and field water losses rather than from low seasonal or annual rainfall totals. A feasible strategy to alleviate this limitation is to harvest excess rainwater in a farm pond during the wet season and use the conserved water for crop production in both wet (as insurance against drought) and dry seasons by adopting suitable crop and cropping systems. The results of water balance in a 1.05 ha field, on which a farm pond was built using 0.09 ha area, showed that 28–37% of seasonal rainfall was available as surface runoff from microcatchment (0.66 ha growing soybean, peanut and pigeonpea) for collection in the pond. This was sufficient for saving rice in a 0.30 ha area (in the lower side of the field) from drought stress, and for establishment of chickpea and mustard (in 0.90 ha) in the post-rainy season after harvest of rainy season crops. Soybean, peanut and pigeonpea, grown in the microcatchment during the rainy season, utilized respectively 371–726, 364–733 and 535–920 mm water in evapotranspiration ($E_t$) and deep percolation ($P$). Rice grown below the pond required 28–317 mm water in different seasons to save the crop from in-season drought stress which commonly occurred during vegetative and reproductive stages. Water requirement ($E_t + P$) of rice was 816–1342 mm in different seasons. Residual soil moisture after rainy season soybean, peanut and rice was sufficient (172–203 mm) to support post rainy season crops of chickpea and mustard. However, the losses of moisture from the soil surface layer after harvest of rainy season crops were rapid (7–23 mm), which necessitated a light irrigation (21–45 mm) for establishment of chickpea and mustard in the post-rainy season. The water balance results of soybean-mustard, peanut-mustard and peanut-chickpea were near identical to soybean-chickpea cropping. Similarly the water balance of rice-mustard was identical to

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rice-chickpea in the vertisols. Soybean-mustard and rice-chickpea were the suitable and economical cropping systems for the microcatchment and service area of the farm pond.

*Keywords:* Field Water balance; Microcatchment; Farm pond; Runoff; Cropping system; Residual soil moisture

## 1. Introduction

Productivity of rainfed areas in eastern Madhya Pradesh, India, is low even though they are endowed with abundant rainfall, well conserved soils and favorable climate that can otherwise support not only a good rainy season crop, but also a crop in the post-rainy season (Ghildyal, 1989). Rainfed rice and rice-based cropping systems are practiced in the region, which suffer from water stress mainly due to uneven distribution of rainfall. Erratic distribution of rainfall and increased field water losses account for unreliability and shortage of water for crop production. Most of the precipitation occurs between June and September. During this period about 50% of the rainfall anywhere in India occurs in about 20–30 h (Pishroty, 1987). This duration hardly accounts for 1% of total rainy season cropping during which the rainfed rice is grown. A large amount of water from these storms is lost by runoff, seepage and percolation ($S$ and $P$). The percolated water recharges the soil profile moisture, but water drained out from the field as surface runoff is lost forever. The solution to water deficiency in rainfed farming in high rainfall areas seems to lie primarily in minimizing non-$E_t$ losses ($S$ and $P$ and runoff) over what is normally attainable with traditional land and water management practices. The collection of excess rainfall runoff in small farm ponds by reducing $S$ and $P$ losses from stored water has been found a suitable option for management of rainwater in vertisols (Pal et al., 1993). Traditionally, rice is grown under a continuously flooded conditions and requires 1500–2000 mm water (Kandaih, 1985). However, the storage capacity of the pond is limited and only life saving irrigation may be provided for rice in the rainy season. Thus, in order to increase resource use efficiency and farmer income, this has to be associated with diversified and intensive cropping by involving suitable deep rooted leguminous and other crops that can make more efficient use of soil moisture, require less of purchased inputs like fertilizer and provide higher economic returns than rice. This paper analyses (i) the on-farm water balance of pigeonpea, soybean, peanut and rice based cropping systems for efficient use of soil moisture and rainfall runoff collected in a pond, and (ii) suitable crops and cropping systems for maximizing productivity of the land.

## 2. Materials and methods

### 2.1. Site and soil

Field investigations were conducted for 4 years (1990–1991 to 1993–1994) at Indira Gandhi Agricultural University, Raipur, India (20°16′N, 81°36′E and 289.6 m a.s.l.).

Water retention profiles of the experimental soil at field capacity and after maximum water extraction by roots (wilting point) are given in Fig. 1 (chickpea). Field capacity was determined in situ 24 h after a thorough wetting, and wilting point (soil moisture

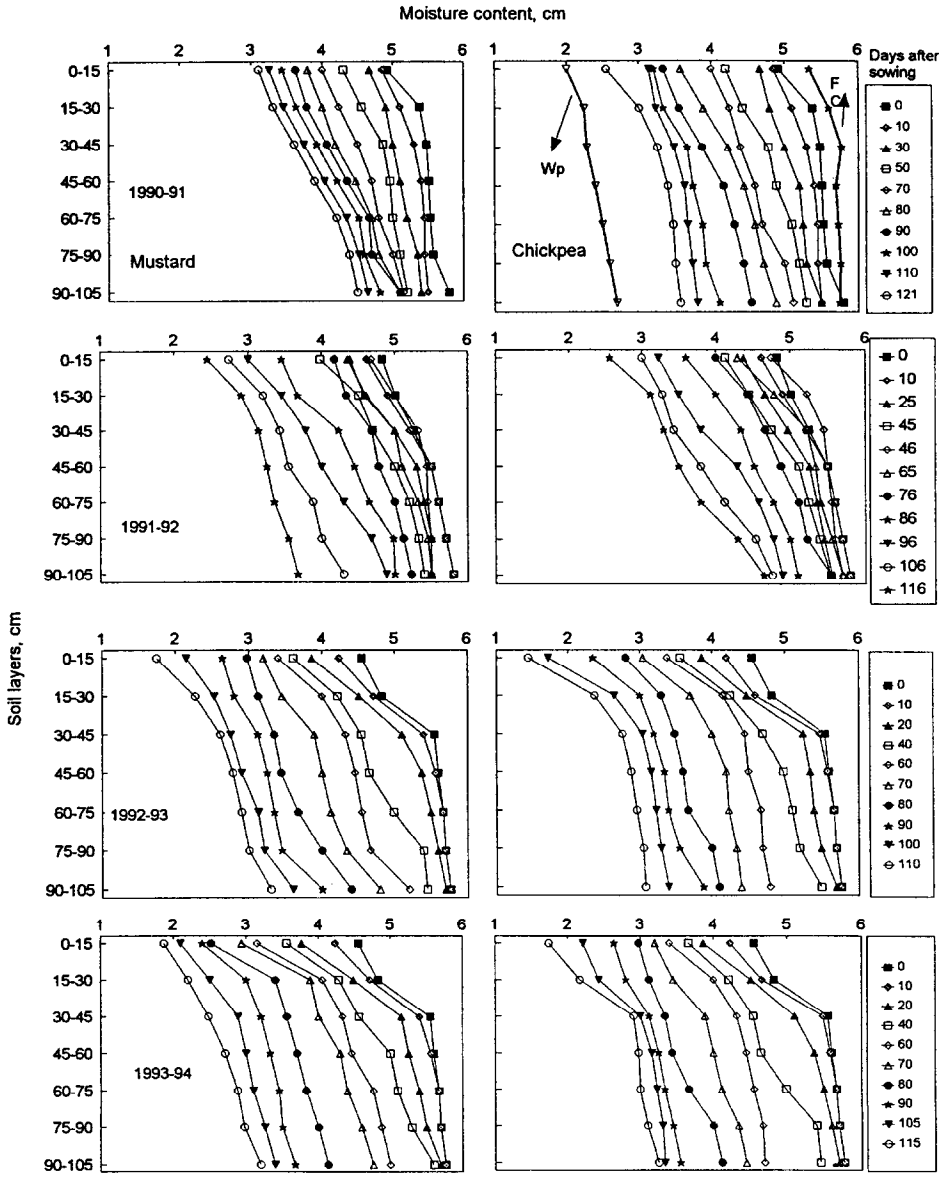



Fig. 1. Soil moisture content at different crop growth periods of mustard and chickpea grown after rice below the pond in different years. Fc-field capacity; Wp-wilting point.

retained at 15 bar suction) was determined using a pressure plate apparatus. Between field capacity (Fc) and wilting point (Wp) the soil retained 244 mm water to a depth of 105 cm. The bulk density increased from the surface to 60 cm depth (1.41–1.68 mg $m^{-3}$), and remained almost constant (1.68 to 1.70 mg $m^{-3}$) to 105 cm soil depth. The soil is neutral in reaction (pH 7.3), low in available nitrogen (288 kg $ha^{-1}$), and phosphorus (8.8 kg $ha^{-1}$) and rich in potassium (428 kg $ha^{-1}$). The field represents deep Vertisols and associated vertic alfisols (locally called Kanhar and Dorsa soils respectively) which cover 45% of the area in the region (Katre, 1989).

### 2.2. Layout plan

A 1.05 ha field was selected during 1990 and a small dugout farm pond of about 750 $m^3$ (enlarged to 970 $m^3$ in 1992) constructed so that about 0.66 ha microcatchment area served for growing upland crops (soybean, pigeonpea and peanut) and for providing runoff to the pond. The pond used about 0.09 ha of the field area. The lower 0.3 ha of the field area is about 30 cm lower than water level at full reservoir capacity so that some irrigation to this area is possible by gravity. The lower 0.3 ha area was used for rice in the rainy season with mustard and chickpea as post-rainy season test crops in the entire field (Fig. 2). The water flow from surrounding nonexperimental fields was cut off by the field bunds and internal drainage trench to avoid runoff from outside getting into the catchment area or the pond. The rainy season crops were sown in the first week of July. Rainfed nursery of rice was grown in the second fortnight of June and transplanted in the second fortnight of July. The rainy season crops were harvested at the end of October; thereafter, mustard and chickpea were sown. All the crops received the recommended fertilizer dose. The growing season of chickpea and mustard lasts from the end of October to first fortnight of March and is characterized by low air temperature and evaporative demand during vegetative and a part of the reproductive phase of crop development.

### 2.3. Field water balance

Field water balance for the cropped area was established using the following relations for the tested cropping systems:

$$dW_s = W_{in} - W_{out} \tag{1}$$

where $dW_s$ is the change in water content of the soil profile for upland crops or change in soil moisture plus change in ponded water depth in field for rice; $W_{in}$ is the sum of input components of water comprising rainwater ($W_{rain}$) and supplemental irrigation ($W_{irr}$) applied from collected runoff (from the cropped area and stored in a small farm pond). $W_{out}$ is the water lost from the field by crop evapotranspiration ($E_t$), deep percolation ($P$) and runoff and/or drainage ($Rd$). Therefore, Eq. (1) can be rewritten as follows, in order to enable the estimation of $E_t + P$, particularly for rainy season crops, for which it is difficult to obtain soil samples for estimating the moisture depletion due to crop uptake, and to measure $P$:

$$E_t + P = W_{rain} + W_{irr} - dW_s - Rd + W_{ro} \tag{2}$$

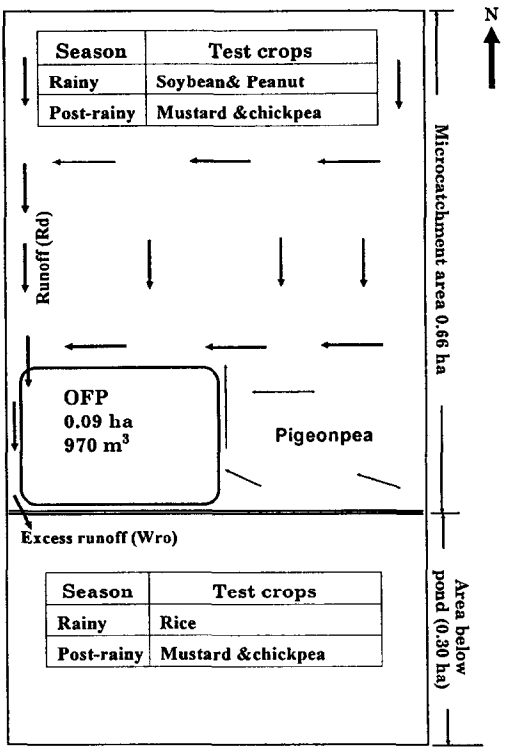



Fig. 2. Schematic lay-out of the field experiment on rainwater management with on-farm pond (OFP). Arrows within the field boundary show directions of runoff water flow in different field segments. The catchment area has a general slope of about 0.5% in north to south and of $< 0.2\%$ in east to west. ←: Slope towards pond contrary to field slope is due to shifted soil.

where $W_{rain}$, $W_{irr}$, $E_t$, $P$ and $dW_s$ are as defined above, $Rd$ is water that was drained out of the field to avoid excessive flooding of the crop and $W_{ro}$ is the excess runoff from the microcatchment after filling the pond. $W_{ro}$ was measured every 12 h by increase in flood water level in the rice field after accounting for water losses due to $E_t + P$. In Eq. (2) the term $W_{ro} = 0$ for non-rice crops grown in the microcatchment, because free surface storage was prevented and runoff from outside the fields was not allowed to enter in the experimental area. The values of $E_t + P$ for soybean, peanut, pigeonpea and rice also include evaporation from bare soil surface after onset of rainfall and before establishment of the crops, and hence represent overestimation. Attempts are being made to work out a reasonable estimate of those evaporation losses which are likely to be within 5 cm, as the crop was established from 5–10 days after the onset of a monsoon. The difference between the moisture balance $(W_{in} - W_{out})$ at the end of the rainy season crop and soil moisture at the start of the post-rainy season crop reflects the water lost between harvest of the first crop and sowing of the next crop.

Initial soil moisture at the beginning of the rainy season was assumed as 180 mm representing the wilting point (at 15 bar suction) for deep soil profile based on separate

estimates for the soil. Rainfall and surface runoff were measured daily and summed for the seasonal totals. Rainfall was measured using a tipping bucket sensor logged to the automatic weather station (AWS) located at the field site. For unrecorded rainfall during a few failures of the AWS and also during the earlier part of the monsoon season when the system was not installed, the rainfall was assumed to be equal to that recorded by the meteorological observatory at the research station. Runoff from the microcatchment was measured by increase in water volume in the farm pond that collected the runoff after accounting for direct contribution of rainfall and losses from the pond during the runoff collection period. Losses due to $E_t + P$ in areas cropped to soybean, peanut and pigeonpea were measured from the difference between (i) rainfall and (ii) loss of water by runoff from the field plus soil profile moisture balance after the crop harvest as given

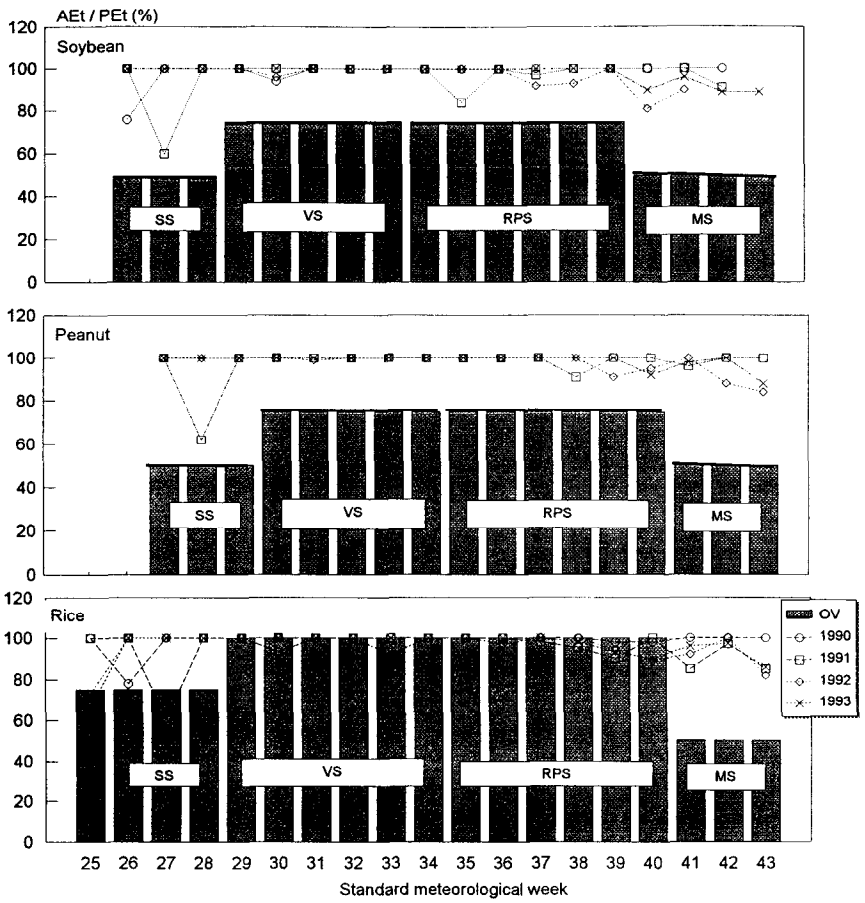



Fig. 3. Agricultural drought during different years in soybean, peanut and rice at different growth stages of the crops. (OV-optimal values of AEt/PEt for the crops, SS-Seedling stage, VS-Vegetative stage, RPS-reproductive stage and MS-maturity stage). The line intercepting the optimal values (OV bars) are the drought stress periods of the crops.

by Eq. (2). Seepage and percolation losses from the field were assumed to be zero for the post-rainy season cropping and no rainfall runoff occurred during that period. $E_t$ for post-rainy season crops was estimated by the relation $E_t = W_{rain} + W_{irr} - d_{Ws}$ which can be obtained from Eq. (2) for the condition: $P = 0$, $Rd = 0$ and $W_{ro} = 0$. The soil moisture depletion ($dW_s$) for post-rainy season crops was estimated by measuring soil moisture depletion in different layers of profile gravimetrically through successive soil samplings at 10/20 day intervals and summing the values to obtain the seasonal total.

For assessing drought or water stress, the ratio of actual evapotranspiration ($AE_t$) to potential evapotranspiration ($PE_t$) was considered (Subrahmanyam et al., 1964; Ramana Rao et al., 1979; Sastri et al., 1981; Reddy, 1983). Weekly values of potential evapotranspiration were computed by the equation of Penman (1948) and $AE_t/PE_t$ values determined by the water balance technique of Thornthwaite and Mather (1955). The optimal values of $AE_t/PE_t$ during seedling, vegetative, reproductive and maturity stages of rice, soybean and peanut are given in Fig. 3 as reported by Patel et al. (1986) for this region.

## 3. Results and discussion

### 3.1. Water balance

The results of water balance are given in Fig. 4, Fig. 5, Fig. 6 and Fig. 7 indicating the availability and use of water for the different crops and cropping systems. The

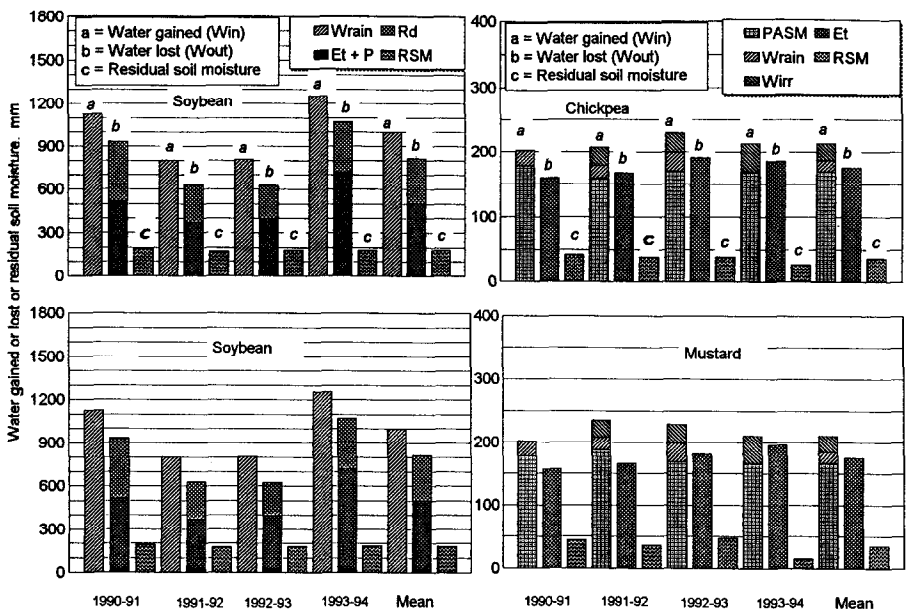



Fig. 4. Field water balance for the cropped area under soybean-chickpea and soybean-mustard in microcatchment of the pond.

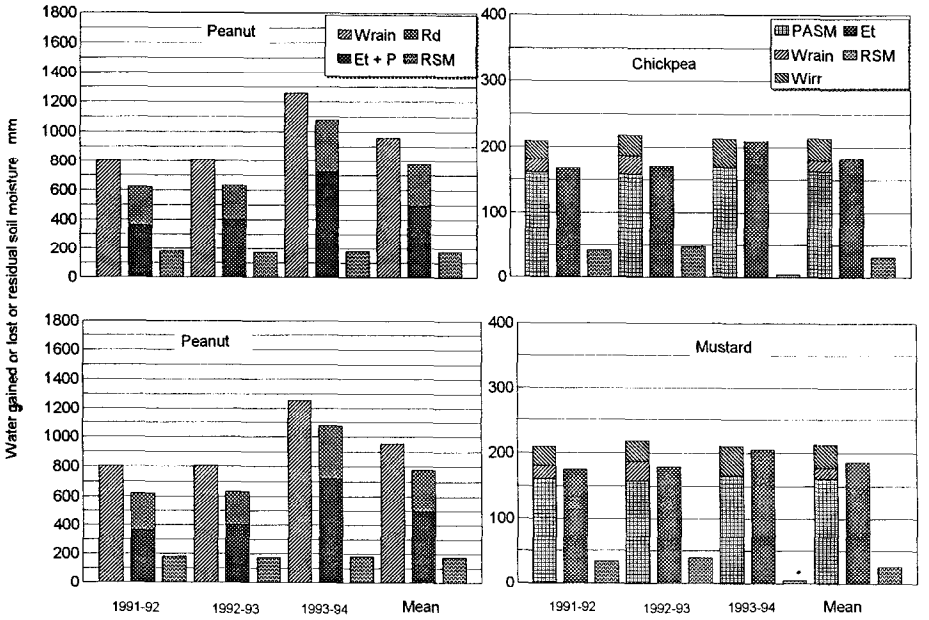



Fig. 5. The water balance components of peanut-chickpea and peanut-mustard cropping in microcatchment of the pond.

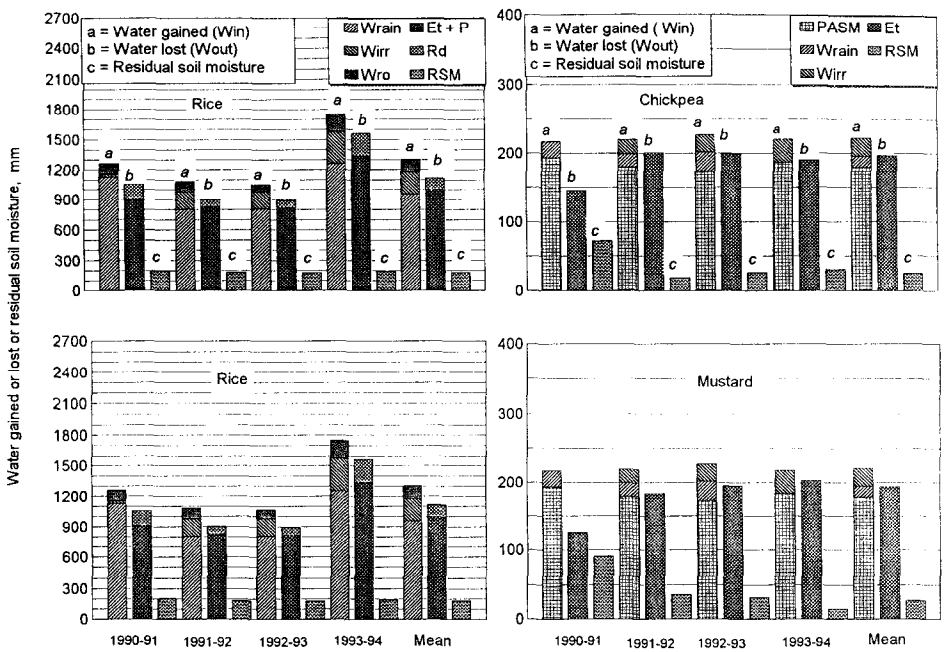



Fig. 6. Field water balance components of rice-chickpea and rice-mustard cropping below farm pond.

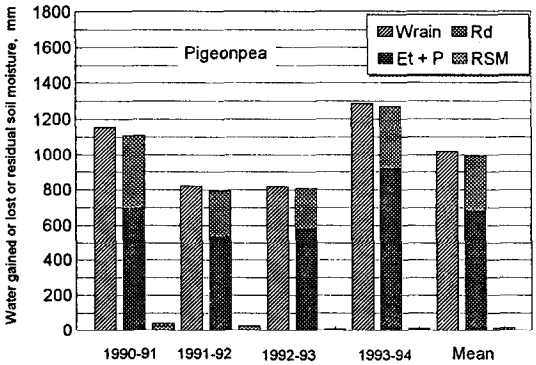



Fig. 7. The water balance components of microcatchment area planted with pigeonpea.

microcatchment received 803 to 1256 mm rainfall in different years during the rainy season. A total of 28 to 37% of rainwater was available as runoff for collection in the pond. Soybean and peanut have identical growth durations. Calculated $E_t$ for soybean and peanut were identical with a maximum difference of 6 mm. The residual soil moisture contents after harvest of the two crops was also identical with a maximum difference of 16 mm between different years. It was therefore assumed that the two crops produced near similar runoff. Pigeonpea has a longer growth duration than the other two crops, but its calculated $E_t$ (average for 4 years) for the time period covered for the above two crops, i.e. for the rainy season, was 11% more than the two crops, and corresponding measured soil moisture level higher by 3%. This supports the assumption that the three crops might have produced near identical runoff. Soybean required 371 to 726 mm water for evapotranspiration and percolation losses ($E_t + P$) depending on the rainfall during crop growth seasons. Estimated $E_t$ of soybean was 336 to 413 mm water, indicating that the percolation losses from the microcatchment area were minimal. The $E_t + P$ in peanut (Fig. 3) was almost similar to that in soybean. Similar findings were reported by Patel et al. (1986). The data indicate that residual soil moisture after rainy season soybean and peanut was sufficient (172 to 194 mm) to support chickpea and mustard in the post-rainy season. However, the loss of moisture from the soil surface layer after harvest of rainy-season crops was quite rapid (7–23 mm) due to which it was difficult to achieve germination and establishment of the chickpea and mustard. The water saved in the pond at the end of the season was used to provide a light supplemental irrigation (28–45 mm) in different years from 1990–1991 to 1993–1994 to achieve satisfactory germination, except in 1990–1991 and 1991–1992 when light rains occurring respectively before and after seeding of the second crop helped to achieve the germination and establishment of the second crop. The $E_t$ of mustard varied in different years from 158 to 209 mm and that of chickpea varied from 160 to 205 mm. The water balance results for soybean-mustard, peanut-mustard and peanut-chickpea were near identical to soybean-chickpea cropping (Fig. 5 and Fig. 6).

Pigeonpea utilized 535–920 mm rainwater for $E_t + P$ in different growing seasons. The estimated $E_t$ for different years ranged from 490 to 568 mm. Thus the percolation

water loss from the pigeonpea field was almost similar as that from soybean field. The crop matured in about 7 months (July to January) and extracted soil profile moisture efficiently as is evident from the low available water left in the soil after harvest of the crop (Fig. 7). This suggested that pigeonpea may perform better than soybean and peanut under water stress conditions.

In the ricefield, 9–17% of the rainwater was not accommodated because deep flooding was not allowed during the early vegetative stage, and also excess water beyond 15 cm depth was drained out during the later vegetative and reproductive stages of rice. In order to increase rainfall efficiency Dhruva Narayan (1982) also suggested storing the rainwater from storms up to 15 cm in the ricefield. Rice utilized 816–1342 mm water for evapotranspiration and deep percolation. The estimated $E_t$ of rice was 473–560 mm water (Fig. 3), indicating that percolation was the major loss from the ricefield, as also reported by Kandaih (1985). The rice in different seasons required 28 to 317 mm water as supplemental irrigation and also benefited from excess runoff from the microcatchment. Residual soil moisture was more (179 to 203 mm) after rice than soybean (172 to 194 mm). Chickpea extracted more water than mustard and showed its suitability under rainfed conditions. The water balance of rice-mustard was identical to rice-chickpea. The water balance analysis also suggested that in order to efficiently use water drained ($Rd$) from the ricefield, the rice area can be increased from an existing 32% to about 40% of the net cultivated area. However, this would require more water for irrigating an extra rice area, which would likely constrain water availability for the establishment of the follow-on crops. A part of the runoff from soybean in excess of farm pond capacity was mainly responsible for the need to drain water from rice when it caused excessive flooding in that crop. Therefore, a more appropriate option seems to increase the capacity of the farm pond by about 100 $m^3$ which is sufficient to accommodate nearly the entire runoff from the microcatchment. This is based on the water harvest index of the pond representing total water volume inflow (runoff from microcatchment + direct rainfall occurring over the open pond area) entering the farm pond relative to its static capacity being 2.7. This would insure increased water availability for rice as well as for post-monsoon crops. Another option is to substitute rice with soybean, but this may have undesirable consequences in terms of increased runoff, increased flooding of the low lying fields, and reduced production of rice.

### *3.2. Water stress and supplemental irrigation*

Soybean and peanut did not suffer from water stress (Fig. 3) during the rainy season and were not given supplementary irrigation. However, initial plant growth of the crops was slightly affected by poor drainage. Pigeonpea was highly susceptible to root zone water logging which caused mortality of the plants. Rice crop experienced drought during most of the growing season, with a slightly higher intensity during 1991 and 1992 than other years (Fig. 3). Drought at the reproductive stage was usual in rice during the four growing seasons. It was also not uncommon for seedling and vegetative stages to experience water stress. This shows that rice under rainfed conditions usually suffers from water stress which may benefit from supplemental irrigation in order to get optimal yield. The rooting system of rice is inherently incapable of efficiently using

moisture from deeper soil layers (Yoshida and Hasegawa, 1982) which explains why rice is so extraordinarily susceptible to short drought periods of 10–15 days. The rice in different seasons required 2–5 irrigations and benefited from excess runoff from the microcatchment area. With the $E_t$ requirement of 5–6 mm day$^{-1}$ that rice needs, a total amount of 5–10 cm water, applied to maintain soil saturation, can effectively carry the crop through a single 10–15 day long dry spell (Pal et al., 1994a). To improve water use efficiency, a limited amount ( < 5 cm) of water was applied at a time in anticipation of rains and to reduce percolation losses.

### 3.3. Crop yields and economic return

Table 1 gives the performance of different crops tested in the Vertisol field with the pond during the rainy and post-rainy seasons in different years. The soybean-chickpea cropping system performed best in the microcatchment followed by the soybean-mustard, peanut-chickpea and peanut-mustard cropping systems in relation to grain yields (Table 1) and net returns (Table 2). Chickpea after soybean suffered from wilt and neck-rot fungal infections, which appear to threaten the sustainability of this system. This is substantiated by the total failure of chickpea due to attacks of wilt and neck-rot in a number of farmers' fields, and none of the available crop varieties are resistant to this disease. Soybean-mustard is therefore an alternative option in that situation. Performance of peanut was poor because the soil being a vertisol, has poor internal drainage and remains wet during the peg formation stage. This created problems in the earthing operation (to cover elongated gynophores for peg formation) leading to less than 50% of the gynophores bearing pods (data not reported here). This phenomenon has also been reported by Rathore and Pal (1994). Pigeonpea under rainfed conditions performed better than peanut and rice, as it exploited profile moisture due to deep rooting and it tolerated drought stress better, but the crop failed under poor drainage conditions which

Table 1
Yield performance of different crops in rainy and post-rainy seasons

| Crop | Grain yield (t ha$^{-1}$) | | | | |
|---|---|---|---|---|---|
| | 1990–1991 | 1991–1992 | 1992–1993 | 1993–1994 | Mean |
| *Rainy season crops* | | | | | |
| 1. Rice | 4.30 | 4.60 | 2.68 | 3.86 | 3.86 |
| 2. Soybean | 2.40 | 2.70 | 3.07 | 2.73 | 2.72 |
| 3. Peanut | NT | 1.30 | 0.81 | 1.11 | 1.09 |
| 4. Pigeonpea | 1.03 | 0.86 | 1.22 | 1.57 | 1.31 |
| *Post rainy season crops* | | | | | |
| 5. Chickpea after soybean | 0.92 | 1.89 | 1.12 | 0.82 | 1.19 |
| 6. Chickpea after peanut | NT | 1.86 | 0.83 | 0.72 | 1.14 |
| 7. Chickpea after rice | 0.86 | 1.41 | 1.04 | 1.19 | 1.13 |
| 8. Mustard after soybean | 0.51 | 1.05 | 0.86 | 0.53 | 0.74 |
| 9. Mustard after peanut | NT | 1.05 | 0.70 | 0.45 | 0.73 |
| 10. Mustard after rice | 0.34 | 0.61 | 0.73 | 0.24 | 0.48 |

NT: Not tried.

Table 2
Economics of different cropping systems in vertisol

| Input cost and income (US $ ha$^{-1}$) | Cropping systems | | | | | | |
|---|---|---|---|---|---|---|---|
| | Rice + Chickpea | Rice + Mustard | Soybean + Chickpea | Soybean + Mustard | Peanut + Chickpea | Peanut + Mustard | Pigeonpea |
| A. Costs: | | | | | | | |
| 1. Input | 190.4 | 175.4 | 217.5 | 202.5 | 216.7 | 200.7 | 100.8 |
| 2. Labour | 220.7(228) | 212.9(220) | 173.2(179) | 165.5(171) | 205.2(212) | 198.4(205) | 104.5(108) |
| 3. Supplemental irrigation | 120.5 | 120.5 | 28.1 | 28.1 | 28.1 | 28.1 | - |
| 4. Total cost (1 + 2 + 3) | 531.6 | 508.8 | 418.8 | 396.1 | 450.0 | 427.3 | 205.3 |
| B. Gross returns [a] | 936.9 | 671.9 | 1177.3 | 983.6 | 864.0 | 663.4 | 433.2 |
| C. Net returns | 405.3 | 163.1 | 758.5 | 587.5 | 414.0 | 236.1 | 227.9 |
| D. Benefit-cost ratio | 1.76 | 1.32 | 2.81 | 2.48 | 1.92 | 1.55 | 2.11 |

[a] Based on average yields as given in Table 1 and price of 1993.Figures given in parentheses are the labour requirement in man days for that system. 1 US $ = 31 Rs.

is a serious problem of deep vertic alfisols and vertisols of the region (Pal et al., 1994b). Though soybean tolerated water logging in the later part of plant growth, at the initial growth period some drainage was required, and was provided by making shallow channels (0.3 m wide and 0.15 m deep) and 5 m wide and 10 m long beds with a wooden plough. The rice-chickpea cropping system performed better than rice-mustard in the service area of the farm pond in terms of grain yield and net returns. Thus soybean-mustard and rice-chickpea appear to be the suitable and sustainable cropping systems. The performance of chickpea and mustard were best in 1991–1992 when light rains which occurred just after sowing enabled good germination of these crops. On account of this, water from the pond was applied as irrigation to the cropped area on the 46th day after sowing, which benefited the crop by maintaining a high soil moisture

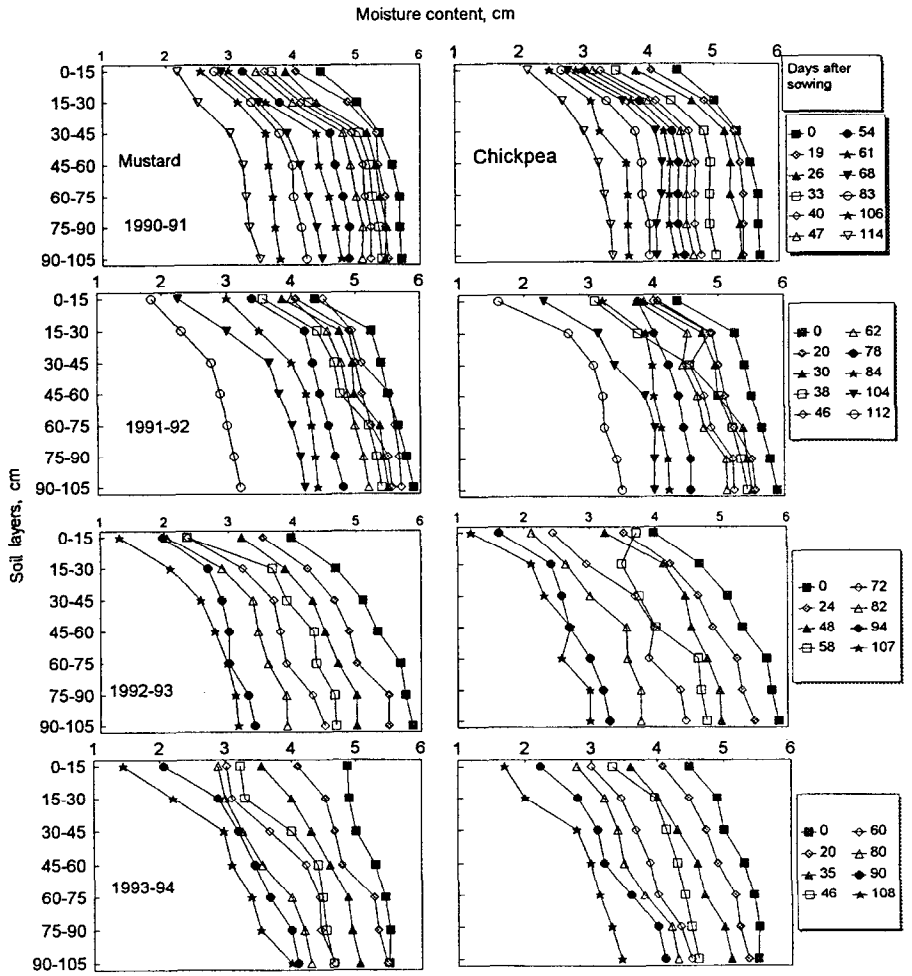



Fig. 8. Soil moisture content during different years at different gorwth periods of mustard and chickpea grown in microcatchment after soybean.

regime in the fertilized soil zone, probably resulting in increased nutrient use efficiency. This advantage was not available in other years. This is evident from Fig. 1 and Fig. 8, which show the moisture regime during growth of chickpea and mustard after rice and soybean. The moisture regimes in the peanut based system were identical to the soybean based system (data not given here). This strongly suggests the need for efficient use of pond water in rice and also for checking the losses of stored water in order to increase water availability, because even one extra irrigation can boost the yield quite substantially.

In this study, the suggested cropping for deep vertic alfisols and vertisols of the region and other similar areas includes substitution of rice mainly with soybean in the microcatchment area above the pond, with the provision of surface drainage, and rice in the remaining third of the field area below the pond. Rice can be saved from drought using water from the pond. With the use of fertilizer and high yielding varieties, rice output from the reduced area was nearly as much as normally obtained by farmers without inputs from the entire field area.

## 4. Conclusions

This study evaluated crop production potential of collection and use of excess rainfall occurring over a field area (1.05 ha) with deep ($> 1$ m) vertisol in the eastern Madhya Pradesh region of India. The test strategy involved substitution of rice with upland crops (viz. soybean, pigeonpea and peanut) in an approximately 0.66 ha area and the construction of a small farm pond (0.09 ha in area) to collect runoff from the field area under upland crops, to save rice from drought in the remaining 0.3 ha field area, and the use of water saved in the farm pond at the end of rainy season for establishing post-monsoon crops after harvest of rice, soybean and peanut. The latter two crops effectively used the soil profile moisture, apparently due to their deeper rooting system, and hence did not face drought, leaving a positive water balance (172 to 194 mm) at the end of the cropping season. The rice crop also left a similarly positive water balance (179 to 203 mm), but required irrigation which the farm pond was able to provide, in order to save it from drought during dry spells, because the crop is inherently incapable of using water from deeper soil layers. Rainfed rice yields are generally less than 1 t $ha^{-1}$ because farmers are discouraged from using fertilizer and other input because of drought. The suggested rainwater management enabled the production of about 1 t rice from a 0.3 ha area as a result of the use of fertilizer in particular, as the collected rainwater provided insurance against drought. Drying of the upper soil layer after the harvest of rice and other upland crops is the major constraint in the establishment of the post-monsoon crops under the traditional system. Harvested rainwater saved at the end of rainy season helped in overcoming the above constraint and hence enabled double cropping. Soybean-mustard and rice-chickpea were the suitable cropping systems for the microcatchment and service area of the farm pond in relation to water use, productivity and net returns.

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